11A — Deep-Dive Feature Audit: Human Foundation, Psychology, Social Graph & Karma (Part 1 of 5)

Genome Generation, 12 Constitutions (`CONST_*`), Spiritual Roots, Curated & 360 Procedural Traits, `data_core.js` $\leftrightarrow$ `human_core.js` Safe Wrapper Layer & Immortal Aging Curves

- **Target Codebase**: `JianghuSect 2` (`JS2-v2.6.0` in Google Drive `Js2 new`)
- **Target Audience**: Autonomous AI Developers (**GPT** & **Gemini**)
- **ABSOLUTE IMMUTABILITY RULE**: **`js/human_core.js` MUST NEVER BE MODIFIED (`HC.FINGERPRINT === '155:2f9c9dd7'`, SHA-256 locked by `test/core_lock.js`)!** Every single fix in Category 11 operates strictly via **`data/data_core.js`** (which is NOT locked) and **external wrapper hooks on `g.HC` methods** inside `js/cultivation.js` / `js/sim.js` (the exact pattern already proven by `HC.bump` / `HC.touch` in `js/stats.js:696`)!
- **Master Roadmap for Category 11 (`11A–11E`)**:
  - **`11A` (This Document)**: **Genome, 12 Constitutions (`CONST_*`), 4-Root Birth System, Curated & 360 Procedural Traits, Broken `data_core.js` Operators/Conditions & Immortal Aging (`bioAge`)**
  - `11B`: Experience Traits (`EXPERIENCE_TRAITS`), Dao Scars (`daoScars`), Voluntary Vows (`VOLUNTARY_VOWS`), Dao Oaths (`daoOath`) & Eight Extraordinary Meridians (`MERIDIANS`)
  - `11C`: Dynamic Psychology — Needs, Mood, Stress, Fear, Trauma, `HC.experience` Drift, Dual Memory Systems & the `heartDemon` / Qi-Deviation Lifecycle
  - `11D`: Social Graph (`rel`), Attachment Styles, Romance, Marriage vs Taoist Companions, Multi-Generational Offspring/Bloodline Inheritance & Mentorship (`master`/`apprentice`)
  - `11E`: Grand Finale — Destiny Karma Threads (`RED`/`GOLD`/`BLACK_THREAD`), `karmicLedger` Sin vs Virtue Halos, Holistic Human-Simulation Synthesis & Master `P0->P3` Roadmap

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1. [HUM-GEN-01] Critical Context Bug (js/human_core.js:286, 295–303 & data/data_core.js): Brand-New 14-Year-Old Mortal Recruits Spawn With Post-Life Acquired Championships, Mahayana Tribulation Scars & Widowed White-Hair Heartbreak!

1.1 Exact Verbatim Code Proof

In `HC.generate` (`human_core.js:286`), birth traits are rolled via `HC.addTrait({ h }, rng, null)`. Inside `HC.addTrait` (`lines 298–300`):

```javascript
const cur = Object.values(C.curatedTraits).concat(C.extraTraits);
const proc = C.procTraits.filter(t => t.tier <= 6);
pool = cur.concat(proc);
```

Because `HC.addTrait` does **not** filter out `category === 'ACQUIRED'` or `category === 'TRAINING'`, **14-year-old Rank-0 mortal farmhands walk through the sect gate at Level 0 already possessing**:

- **`TR_AC_TOURNAMENT_CHAMPION`** (*"เจ้ายุทธจักรผู้ชนะเลิศมหาประลอง — ชื่อเสียงและบารมีเป็นที่ยอมรับทั่วเก้าทวีป"*) while `fame = 0` and `sparWin = 0`!
- **`TR_AC_TRIBULATION_SCAR`** (*"รอดชีวิตจากฟ้าทัณฑ์เก้าสายฟ้า"*) before ever entering Stage 1 (`9-bolt tribulation only occurs at Stage 7+`)!
- **`TR_AC_HEART_DEMON_LOVE`** (*"สูญเสียคนรักไปในศึกยุทธภพจนเส้นผมกลายเป็นสีขาวโพลน"*) at age 13 with `spouse: null` and jet-black hair (`#1a1a22`)!
- **`TR_AC_MASTER_VENGEANCE`** (*"ผู้แบกหนี้เลือดล้างแค้นให้อาจารย์"*) with `master: null`!
- **`TR_AC_BETRAYER_BRAND`**, **`TR_AC_ENLIGHTENED_BODHI`**, **`TR_AC_CLIFF_SURVIVOR`**, **`TR_AC_SEVERED_MERIDIAN`**, plus **`120 category: "ACQUIRED"` procedural traits**!

1.2 Zero-Core-Edit Fix (`External Wrapper on HC.addTrait in js/cultivation.js`)

Notice line `286` of `human_core.js` invokes **`HC.addTrait({ h }, rng, null)`** via property lookup on `HC`!

- Wrap `HC.addTrait` in `js/cultivation.js`: when `id == null && !npc._allowAcquiredRoll`, draw exclusively from `category ∈ ['BLOODLINE', 'INNATE', 'UPBRINGING']`!

- Then award the **8 curated `TR_AC_*` traits organically from real simulation events**:
    1. `TR_AC_TOURNAMENT_CHAMPION` $\leftarrow$ winning Grand Jianghu Assembly (`ELDER_PEAK` bracket) or 3× Sect Tournaments!
    2. `TR_AC_TRIBULATION_SCAR` $\leftarrow$ surviving a 6-bolt or 9-bolt `CULT.TRIBULATION` (`BACKFIRE` or `< 20%` HP survival)!
    3. `TR_AC_HEART_DEMON_LOVE` $\leftarrow$ when an NPC's `spouse` / `companionId` is slain in combat (`AI.kill` — **also turning `npc.h.body.hair = 'ขาวหิมะ'; npc.h.body.hairHex = '#e9edf5'` to match the lore text!**)!
    4. `TR_AC_MASTER_VENGEANCE` $\leftarrow$ when an NPC's `mentor` / `mentorId` is slain by a hostile killer!
    5. `TR_AC_SEVERED_MERIDIAN` $\leftarrow$ catastrophic breakthrough failure or crippling duel defeat!
    6. `TR_AC_CLIFF_SURVIVOR` $\leftarrow$ near-death expedition survival event (`lucky cliff cave encounter`)!
    7. `TR_AC_ENLIGHTENED_BODHI` $\leftarrow$ overcoming `heartDemon >= 80` down to `0` via Bodhi meditation / Samsara trial!
    8. `TR_AC_BETRAYER_BRAND` $\leftarrow$ recruited defector (`RECRUIT_DEFECTOR`) or failed mutineer!

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2. [HUM-MOD-02] Broken Operators, Ignored `condition` Clauses & Non-Existent Stat Paths in `data/data_core.js`!

Cross-checking every modifier in `data/data_core.js` against `HC.traitMods` (`human_core.js:112–117`) and `HC.STAT_DEFS` reveals **four silent bugs**:

2. **`op: "MAX_CAP"` Interpreted as `×62` Multiplier on `TR_IN_CRIPPLED_ROOT` (`01A` Confirmed)**:
    - `{"stat":"aptitudes.cultivation","op":"MAX_CAP","value":62}` enters the `else` branch of `HC.traitMods`: `{ op: 'MULT', value: 1 + (62 - 1) * k }` = **`×62` multiplier**, clamping every "Crippled Root" disciple's `aptitudes.cultivation` to **`100` (`God-tier Qi genius`)**!
    - **Fix**: Replace `"MAX_CAP"` in `data_core.js` with `{"stat":"aptitudes.cultivation","op":"MULT","value":0.55}` (and enforce the `<= 62` cap in a lightweight `HC.eff` wrapper or `CULT.pathScores`)!
2. **Dead Non-Existent Stat Path `physiology.poisonResistance` on `TR_BL_POISON_IMMUNE`**:
    - `TR_BL_POISON_IMMUNE` (กายาหมื่นพิษไม่กล้ำกราย) grants `{"stat":"physiology.poisonResistance","op":"ADD","value":65}`

—**`physiology.poisonResistance` does NOT exist in `HC.STAT_DEFS` (`the real stat in human_core.js:48 is physiology.immunity`)**!
    - Furthermore, its description promises *"กลืนกินสมุนไพรพิษมาเพิ่มพลังปราณแทนโอสถได้โดยไม่เป็นอันตราย"*, yet holders take `100%` poison DoT in `combat.js` and `100%` `c.toxic` from pills!
    - **Fix**: Map to `"physiology.immunity"` (`+35`) in `data_core.js`, reduce `poison`/`venomSeed` DoT & `c.toxic` gain by `-75%` when `hasTrait('TR_BL_POISON_IMMUNE') || constitution === 'CONST_POISON_SAC'`, and let `CONST_POISON_SAC` convert resisted poison ticks into `+3% MaxQi`!
3. **Blind Swordsman (`TR_IN_BLIND_SWORD_EAR`) Has Normal `vision = 75` & Still Gets Blinded by Flashbangs!**:
    - `TR_IN_BLIND_SWORD_EAR` (*"สูญเสียการมองเห็นแต่กำเนิด แต่ไม่ได้รับผลกระทบจากวิชาลวงตา"*) never reduces `physiology.vision` (`add physiology.vision -80, physiology.hearing +28` in `data_core.js`!) and `combat.js` / `ST.addStatus` never checks `unlockTags: ['ILLUSION_IMMUNE']`—so a **congenitally blind swordsman gets afflicted by `'blind'` (ตาพร่ามัว)**! Block `'blind'` status application when `hasUnlockTag(u, 'ILLUSION_IMMUNE')`!
4. **Conditional Trait Modifiers (`condition: { maxHealth, requiredSeason, inCombatAgainstGrudge }`) Fire Unconditionally 24/7**:
    - `human_core.js` ignores `m.condition`, so `TR_BL_BLOOD_CURSE` (`1.18× maxHealth: 65`), `TR_AC_MASTER_VENGEANCE` (`1.22× inCombatAgainstGrudge: true`), `TR_BL_PHOENIX_FLAME` (`+10%` BT at `maxHealth: 50`), and `TR_BL_YIN_BROKEN_VEIN` (`-15` recovery in `WINTER`) apply permanently even at full HP, in summer, and against innocent sparring partners! Move conditional portions out of static `modifiers` into `stats.js` / `combat.js` / `cultivation.js` runtime checks!

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3. `[HUM-CONST-03]` All 12 Constitutions (`CONST_*`) Have `"element": "NEUTRAL"`, Zero `toughness`/`vitality`/`strength` Scaling & Unwired Lore Effects (`data/data_core.js`)!

1. **Wrong `"element": "NEUTRAL"` on Elemental Constitutions**:
    - `CONST_FLAME_CAULDRON` (`FIRE`), `CONST_WOOD_SPRING` (`WOOD_LIFE`), `CONST_YIN_FROST_VEIN` (`WATER_ICE` / `YIN_SHADOW`), and `CONST_STAR_MIND` (`ASTRAL` link) are all marked `"element": "NEUTRAL"` (`EXPLICIT_NON_ELEMENTAL`), denying them elemental resonance!

2. **Missing `physiology.toughness`, `vitality` & `strength` Across All Body Constitutions**:
    - `CONST_IRON_VAJRA`, `CONST_BEAST_BLOOD`, and `CONST_DRAGON_MARROW` boost `physiology.endurance`, **but grant `+0` `physiology.toughness`, `+0` `physiology.vitality`, and `+0` `physiology.strength`**—even though `HC.derived(npc).hpMax` (`human_core.js:211`) and `ST.compute def` (`stats.js:273`) scale primarily off **`vitality` and `toughness`**! Add `toughness` and `vitality` modifiers in `data_core.js`!
3. **Unwired Constitution Passives**:
    - Wire `CONST_IRON_VAJRA` & `TR_BL_VAJRA_BONE` & `TR_AC_TRIBULATION_SCAR` (`cultivation.tribulationResist`) into `CULT.TRIBULATION` lightning mitigation!
    - Wire `CONST_WOOD_SPRING` (`2×` daily passive `c.toxic` natural purge), `CONST_STAR_MIND` (`ASTRAL` special access), and `CONST_BEAST_BLOOD` (`BLOOD_DEMON` access + `+18%` `attemptTameBeast` chance)!
4. **Procedural Trait (`TR_GEN_*`) Naming Typos (`data/data_core.js`)**:
    - Fix duplicated Thai substrings like `TR_GEN_3_19` (`"ดวงจิตโกลาหลบรรพกาลบรรพกาลเทวะ"`) and contradictory curse titles like `"รอยแผล...สวรรค์ประทาน"` / `"ปมในใจ...ระดับเซียนอมฤต"` where auspicious suffixes were concatenated onto `isCurse: true` traits, and populate the empty `GENERAL` prefix group (`39%` of `procTraits`) with distinct secondary modifiers!

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4. `[HUM-AGE-04]` Premature Senility Curve (`bioAge > 30`) on High-Realm Immortals & Unsynced `npc.lifespanMax` (`js/human_core.js:140–152`)!

1. **`npc.lifespanMax` Desyncs From Mantra & Pill Lifespan Extensions**:
    - `HC.bioAge(npc)` (`human_core.js:142`) reads `npc.lifespanMax || 80`. Whenever `EVERGREEN_LONGEVITY_SPRING` (`+28%` lifespan), `BIG_DIPPER_LONGEVITY_SAGE`, or `EFX_LIFESPAN_EXTEND` modifies max lifespan without updating `npc.lifespanMax` + calling `HC.touch(npc)`, `HC.bioAge` computes inflated biological age! Centralize `CULT.syncLifespan(npc)` to always update `npc.lifespanMax` identically.
2. **Stage 7–9 Immortals Begin Physical & Mental Senile Decay at Just `38%` of Their Lifespan (`bioAge = 31`)**:
    - Because `AGE_CURVES.P` and `AGE_CURVES.F` start decaying at `bio > 30`, a Stage-9 Immortal (`lifespanMax = 2,000` years) begins losing Strength, Speed, Reflex, Dexterity, Reasoning, and Focus at age `760` (`with 1,240 years still to live!`).

- **Zero-Core-Edit Fix**: Wrap `HC.ageFactor = (kind, bio) => ...` in `js/cultivation.js` (or apply a cultivation realm vitality floor when `equivalentStage >= 5` so immortals remain at `1.00×` prime until `bio >= 68` — the authentic Xianxia *Five Decays of Heaven and Man* 天人五衰 twilight window!).